

RUSHIKESH MOHALKAR

AI/ML Engineer

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PROFESSIONAL SUMMARY

Results-driven AI/ML Engineer with hands-on experience designing scalable ML systems, optimizing LLM pipelines, and deploying production-ready generative AI solutions. Proficient in reinforcement learning, deep learning architectures (CNNs, RNNs, LSTMs), and multimodal systems. Demonstrated track record delivering high-accuracy medical classifiers (92–95%), robust RL agents (+40% track completion), and optimized QA workflows (20–30% efficiency gains). Seeking to build high-impact AI products in fast-paced engineering environments.

PROFESSIONAL EXPERIENCE

Programmer Analyst — Cognizant

July 2024 – Present

- Executed functional and regression testing across enterprise systems, improving defect detection rate by ~30% through structured test planning and systematic coverage.
- Authored 2K detailed test cases and collaborated cross-functionally with engineering teams, reducing release-cycle delays by 20%.
- Optimized QA reporting workflows by building automated defect-analysis pipelines, cutting test execution time by 15% and accelerating delivery cycles.

EDUCATION

B.E. — Electronics & Telecommunications Engineering AISSMS College of Engineering, Pune (affiliated with Savitribai Phule Pune University)

TECHNICAL SKILLS

Languages: Python, Java, SQL, Bash

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers

Generative AI & LLMs: LangChain, LlamaIndex, RAG Pipelines, Prompt Engineering, QLoRA, Ollama

NLP: NLTK, Named Entity Recognition, Text Classification, Sequence Modeling

Cloud & DevOps: AWS (S3, EC2, SageMaker), Docker, Kubernetes, FastAPI, Flask, Git, Linux

Core Competencies: Reinforcement Learning, Transfer Learning, Model Optimization, Data Preprocessing, Time-Series Forecasting

PROJECTS

Oryza Chatbot — Self-Hosted LLM Application

- Developed a self-hosted AI chatbot using Flask and Ollama running the Qwen 2.5 1.5B LLM locally, enabling private, offline conversational AI with zero cloud API dependency.
- Containerized the full application stack using Docker and Docker Compose with Nginx as a reverse proxy, achieving a production-ready, scalable deployment architecture.

Autonomous Self-Driving Car using PPO Reinforcement Learning

- Built a Pygame-based autonomous driving agent using Proximal Policy Optimization (PPO) with 17-dimensional observation space, domain randomization, and continuous/discrete action spaces — improving track-completion consistency by 40% and reducing collision rate by 35%.
- Engineered advanced reward shaping, physics-driven vehicle dynamics, and multi-track simulation pipelines, achieving 25% smoother cornering, faster policy convergence (~100K steps), and robust generalization to unseen tracks.

Pneumonia Detection using Transfer Learning (VGG16)

- Developed a VGG16-based transfer learning model for chest X-ray binary classification, achieving 92–95% accuracy with strong generalization via data augmentation and optimized preprocessing.
- Reduced inference latency by 25% through lightweight top-layer design and efficient pipeline optimization, enabling fast, reliable NORMAL vs. PNEUMONIA diagnosis on unseen medical images.

Lung Cancer Detection using Convolutional Neural Networks

- Engineered CNN-based models for CT-scan lung cancer detection, achieving 89%+ precision and significantly reducing false positives through optimized preprocessing and augmentation strategies.
- Enhanced diagnostic recall by 12% using noise-reduction and contrast-enhancement pipelines; accelerated training by 20% via targeted hyperparameter tuning.

Movie Recommender System using Restricted Boltzmann Machines (RBMs)

- Built an RBM-based collaborative filtering system trained on 1M+ user-movie interactions using CD-10 and binary implicit feedback to learn latent preference factors, achieving a test MAE of ~0.25.
- Engineered a full data pipeline (943×1682 user-item matrix) with optimized Gibbs sampling and training loops, improving recommendation accuracy by ~15%.

Google Stock Price Forecasting using LSTM RNNs

- Designed and trained a multi-layer LSTM time-series forecasting model achieving <0.02 MSE, accurately capturing Google stock price trends with dropout regularization for generalization.
- Built a robust preprocessing pipeline including MinMax scaling, 60-day lookback windowing, and sequence reshaping — enhancing prediction stability by 25% on held-out test data.

CERTIFICATIONS

- [Mathematical Foundations of Machine Learning](#)
- [Deep Learning A–Z: Neural Networks, AI & ChatGPT](#)
- Generative AI with LangChain & Hugging Face (In Progress)

[Python for Data Science & Machine Learning Bootcamp](#)

AI Engineer Core Track: LLMs, RAG, QLoRA, Agents (In Progress)